PALLIATIVE CARE RESEARCH COOPERATIVE GROUP

PCRC De-identified Data Repository (DiDR) Study Summary

TITLE: **Pilot Testing a Virtual Reality Protocol for Improving Pain and Pain-Related Distress in Patients with Advanced Stage Colorectal Cancer**		
PRINCIPLE INVESTIGATOR(S):	**Sarah A. Kelleher, PhD**	SITE(S) (if applicable): **N/A**
COORDINATING SITE:	**Duke Cancer Institute**	
STUDY PERIOD		
START:	**March 12, 2020**	
LAST SUBJECT CONTACT:	**July 19, 2021**	

OBJECTIVES:

Aim 1: Examine the feasibility, acceptability and safety of a 30-minute VR Blue session in patients with advanced colorectal cancer. Hypothesis: Feasibility will be determined by accrual (N=20 patients over 6 months), >80% adherence to the protocol (defined as the degree to which participants are willing and able to complete the 30-minute VR exposure), and >80% completed data (measured by percent data collected at the study appointment, including pre-, midpoint, and post-VR assessments). Acceptability will be demonstrated by 80% of participants reporting VR satisfaction on the Client Satisfaction Questionnaire (mean score of 7/8). Safety of the VR protocol will be demonstrated by 80% of participants completing the VR session without significant side effects, such as motion sickness, dizziness, headache, and nausea.

Aim 2: Collect initial data on the impact of VR Blue on pain, tension and distress, and describe the relationship between pre- to post-VR changes in cognitive variables (i.e., pain catastrophizing, pain self- efficacy) and changes in pain, tension and distress over the course of the VR session. Hypothesis: 2A. Reductions in pain, tension and distress will be demonstrated during the VR session in colorectal cancer patients. 2B. Pre- to post-VR reductions in pain catastrophizing and increases in pain self-efficacy will be correlated with improvements in pain, tension and distress occurring over the course of the VR session.

Aim 3: Gather qualitative post-session data on patients' experiences, thoughts and feelings about VR to optimize VR Blue for a future randomized clinical trial testing a multi-session VR protocol. Hypothesis: This data will lead to optimization and manualization of a multi-session VR Blue protocol to manage pain in advanced colorectal cancer patients that can be tested in a larger, randomized clinical trial.

PARTICIPANTS		
	ENROLLMENT	ELIGIBILITY CRITERIA
Patients:	**20**	**Eligibility criteria include: a) age 18-85, b) stage IV cancer diagnosis, c) moderate clinical pain documented in medical chart (≥4 on 0-10 scale) and confirmed on the day of their study appointment (≥4/10), d) ability to read and speak English, e) self-reported normal or corrected to normal vision, and f) self-reported normal hearing.**
Informal Caregivers:	Click here to enter text.	**N/A**
Health Care Providers:	Click here to enter text.	**N/A**

METHODOLOGY:

Procedures. Study Appointment. The study appointment will take place at the Duke Pain Prevention and Treatment Research Program office. At this appointment, research staff will explain study procedures and conduct informed consent. Once informed consent is obtained, patients (N=20) will complete the pre-VR assessment (<30 min) via REDCap, a HIPAA/PDPA compliant, web-based survey system. Following completion of this assessment, participants will complete the VR exposure and the midpoint assessment. Participants will then complete the post-VR assessment immediately following VR Blue exposure. Participants will be compensated $40 for completing this study appointment. Participants will continue to receive usual care and will not be asked to change or decline any strategies for pain management.

Virtual Reality Exposure. We will gather preliminary data on participants' response to a single 30-minute laboratory-based virtual reality underwater/sea environment (VR Blue) session. VR Blue is an immersive computer-generated environment featuring calming scenic graphics and relaxing nature music. VR Blue has been shown to increase tolerance for thermal pain stimuli in healthy participants by Dr. Colloca, consultant on this application. In the proposed study, a member of the study team will familiarize the participant with the VR glasses and screen, and guide them through a demonstration of the VR environment. Participants will be oriented to the VR's video and auditory stimuli with an emphasis on becoming relaxed and fully immersed in the virtual environment. Participants will be instructed to pay close attention to the virtual environment around them to cultivate their awareness of what they see. Following this orientation, the participant will complete the 30-minute VR exposure. At the midpoint, participants will be asked to provide ratings of pain, tension and distress.

INTERVENTION (if applicable):

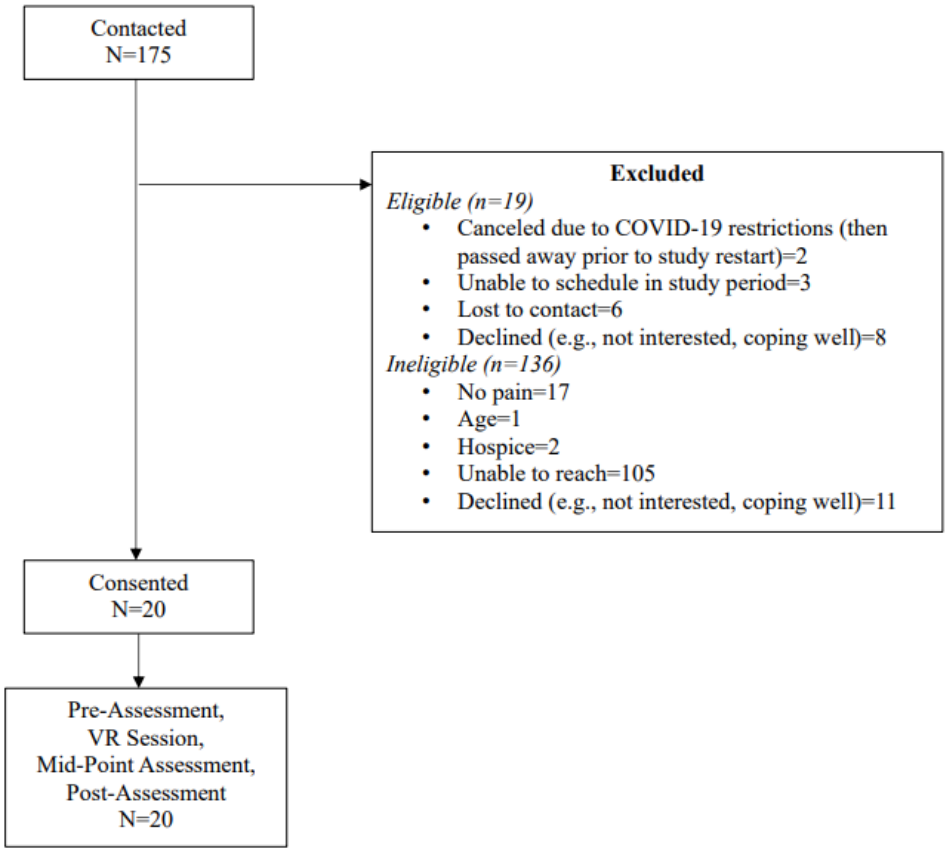
N/A

MEASURES:

Study Measures. Aim 1: Feasibility will be assessed by examining (a) study accrual (N=20 over 6 months), (b) protocol adherence (>80% adherence to the protocol [defined in this study as the degree to which participants are willing and able to complete the 30-minute VR exposure]), and (c) completed data (>80% data collected at the study appointment, including pre-, midpoint, and post-VR assessments). Acceptability will be assessed post-VR using the Client Satisfaction Questionnaire.31 Safety of the VR protocol will be assessed based on participants' report of VR side effects, such as motion sickness, dizziness, headache, nausea, or other negative physical reactions. Aim 2: Pre-, midpoint- and post-session ratings of pain, tension, stress, anxiety, and mood. Pain. Pain will be assessed using the Brief Pain Inventory (BPI)32 measuring pain severity and interference. This measure has been widely used in cancer patients. Tension, Stress, Anxiety and Mood. Tension, stress, anxiety and mood will be measured with Visual Analogue Scale (VAS) items with ratings ranging from 0 to 100. The timeframe reference will be "right now". VAS's for measuring tension, stress, anxiety and mood have been used in prior VR studies. Pre- and post-session measures of expectancy and enjoyment. We will use Visual Analogue Scale (VAS) items with ratings ranging from 0 to 100 to measure how much participants expect to improve from participating in the VR Blue intervention (pre) and how much they enjoyed participating in the VR Blue intervention (post). Pre- and post-session measures of pain catastrophizing and self-efficacy for pain. Pain Catastrophizing. Pain catastrophizing will be measured with the Coping Strategies Questionnaire's Pain Catastrophizing subscale.33,34 The timeframe reference will be "right now". Items will be rated on a scale ranging from 0=never do to 6=always do when in pain. Pain Self-Efficacy. Self-efficacy will be measured using items adapted from the self-efficacy for pain management subscale of the Chronic Pain Self-Efficacy Scale.35 For example, participants will be asked, "How confident are you

that you could use virtual reality to decrease your pain?" Items are rated on a 10-point scale ranging from 10=very uncertain to 100=very certain. Scores are averaged to give an overall value for pain self-efficacy. Aim 3: We will conduct exit interviews with participants to collect qualitative data to refine and improve the VR protocol.

SUBJECT FLOW (CONSORT):



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STUDY CALENDAR:

Date	October 8, 2021																										
PI	Sarah Kelleher, PhD																										
TITLE	Pilot Testing a Virtual Reality Protocol for Improving Pain and Pain-Related Distress in Patients with Advanced Stage Colorectal Cancer																										
		Year 1												Year 2													
	Pre	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19	M20	M21	M22	M23	M24	Post1	Post 2	Total
		J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	
Project Start Up																												
Participant Recruitment										2	COVID-19 PANDEMIC RESTRICTIONS																	
Intervention and assessments																												
Data Prep and analysis																												
Manuscript and grant writing																												
Planned accrual=20																												
Target Accrual					3	3	3	4	4	3	COVID-19 PANDEMIC RESTRICTIONS																	20
Revised Target Accrual										2										1	2	5	5	5				20
Actual Accrual										2										2	2	1	1	5	6	1		20

BASELINE CHARACTERISTICS (TABLE 1)

Table 1. Participant Demographic Characteristics (N=20)

	N (%)	M	SD	Range
Age (years)		56.66	10.73	32-80
Gender				
Male	14 (70%)			
Female	6 (30%)			
Race				
White	15 (75%)			
Black or African American	4 (20%)			
Native Hawaiian or Pacific Islander	1 (5%)			
Ethnicity				
Non-Hispanic or Latino	18 (90%)			
Hispanic or Latino of any race				
Declined to Report				
Unknown	1 (5%)			
Other	1 (5%)			
Education				
Less than HS	1 (5%)			
HS Diploma	3 (15%)			
Some College	4 (20%)			
Bachelor's Degree	4 (20%)			
Graduate Degree	8 (40%)			
Marital Status				
Never Married	2 (10%)			
Married	16 (80%)			
Widowed	2 (10%)			

Note. *M* = mean; *SD* = standard deviation; HS = High school.

PCRC STANDARDIZED DATA ELEMENTS

Please see the separate information sheet "DISC Standardized Data Elements" for the exact wording and format of the data elements.

DATA ELEMENT	Collected?	Var Name(s)	Data source (e.g. self-report, EHR) or reason not applicable
1. Site ID (if multi-site)	☐		N/A – one-site
2. Who is the research participant? (e.g., patient, caregiver, etc.)	☒	MRN	EHR
3. Sex	☒	Sex	EHR
4. Ethnicity	☒	Ethnicity	Self-report
5. Race	☒	Race	Self-report
6. Age in years	☒	Age	Self-report
7. Current Marital Status	☒	Marital	Self-report
8. Primary life-limiting diagnosis/illness	☒	Cancertype	EHR
9. Performance status (AKPS)	☐		N/A, not collected
10. Enrolled in Hospice	☐		N/A, not collected, hospice, exclusion criteria
a. If yes to hospice, where is hospice care provided?	☐		
11. Receiving Palliative Care (PC)?	☐		N/A, not collected
a. If yes to receiving PC, where is PC provided?	☐		
12. Source of Death information	☐		N/A, not collected
13. Location of Death	☐		N/A, not collected, must be alive for study
14. Enrolled in Hospice at time of death?	☐		N/A, not collected, must be alive for study
15. Receiving PC at time of death?	☐		N/A, not collected, must be alive for study

Cells in blue only need to be collected for patient research participants. Cells in orange should be collected regardless of participant type.

PATIENT REPORTED OUTCOME INSTRUMENTS

CONTENT *(e.g., PS)*	ABBREV *(e.g., AKPS)*	INSTRUMENT NAME *(e.g., Australian Modified Karnofsky Performance Status)*
BPI	BPI	Brief Pain Inventory – Pain Severity
BPI	BPI	Brief Pain Inventory – Pain Interference
BPI	BPI	Brief Pain Inventory - Medications
VAS	VAS	Visual Analog Scale
SE	SE	Self-Efficacy for Pain Control
PC	PC	Pain Catastrophizing (Subscale of Coping Strategies Questionnaire)
CSQ	CSQ	Client Statisfaction Questionnaire